

500 Curated Easy DSA Coding Problems

Database-oriented beginner problem bank. Each problem has a unique slug, complete statement, constraints, I/O format, three visible tests, hidden tests, tags, and complexity target.

1. Find the Largest Element

Slug: find-the-largest-element-001 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Given an integer array, print its largest element.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

2. Find the Smallest Element

Slug: find-the-smallest-element-002 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Given an integer array, print its smallest element.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 1

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: -4

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -1000000000

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

3. Sum of Array Elements

Slug: sum-of-array-elements-003 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Calculate the sum of all elements in an integer array.

Constraints: $1 \leq N \leq 10^5$; values fit in 32-bit signed integers; sum fits in 64-bit

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

4. Count Positive Values

Slug: count-positive-values-004 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many elements of an array are strictly positive.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

5. Count Negative Values

Slug: count-negative-values-005 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many elements of an array are strictly negative.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

6. Count Even Values

Slug: count-even-values-006 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count the even integers in an array.

Constraints: $1 \leq N \leq 10^5$; values fit in 32-bit signed integers

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

7. Reverse an Array

Slug: reverse-an-array-007 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Reverse the array in place and print the resulting sequence.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

8. Print Alternate Elements

Slug: print-alternate-elements-008 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print elements at indices 0, 2, 4, and so on.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

9. Find the Second Largest Distinct Value

Slug: find-the-second-largest-distinct-value-009 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print the second largest distinct element, or -1 if it does not exist.

Constraints: $2 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

10. Check if Array Is Sorted

Slug: check-if-array-is-sorted-010 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Determine whether the array is non-decreasing.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

11. Move Zeros to the End

Slug: move-zeros-to-the-end-011 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Move all zero values to the end while preserving the order of non-zero values.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

12. Count Occurrences of a Target

Slug: count-occurrences-of-a-target-012 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many times X occurs in the array.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

13. Find First Occurrence

Slug: find-first-occurrence-013 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print the first index of X, or -1 if absent.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

14. Replace Negative Values

Slug: replace-negative-values-014 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Replace every negative element with zero and print the array.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

15. Difference Between Maximum and Minimum

Slug: difference-between-maximum-and-minimum-015 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print $\max(A) - \min(A)$.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

16. Left Rotate by One

Slug: left-rotate-by-one-016 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate all elements one position to the left.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

17. Right Rotate by One

Slug: right-rotate-by-one-017 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate all elements one position to the right.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

18. Left Rotate by K

Slug: left-rotate-by-k-018 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate the array left by K positions.

Constraints: $1 \leq N \leq 10^5$; $0 \leq K \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

19. Check Majority of Ones

Slug: check-majority-of-ones-019 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Determine whether ones occur more often than zeros in a binary array.

Constraints: $1 \leq N \leq 10^5$; $A[i]$ is 0 or 1

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

20. Find Missing Number

Slug: find-missing-number-020 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: An array contains distinct values from 0 through N with one missing. Find it.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

21. Find the Largest Element

Slug: find-the-largest-element-021 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Given an integer array, print its largest element.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

22. Find the Smallest Element

Slug: find-the-smallest-element-022 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Given an integer array, print its smallest element.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

23. Sum of Array Elements

Slug: sum-of-array-elements-023 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Calculate the sum of all elements in an integer array.

Constraints: $1 \leq N \leq 10^5$; values fit in 32-bit signed integers; sum fits in 64-bit

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

24. Count Positive Values

Slug: count-positive-values-024 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many elements of an array are strictly positive.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

25. Count Negative Values

Slug: count-negative-values-025 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many elements of an array are strictly negative.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

26. Count Even Values

Slug: count-even-values-026 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count the even integers in an array.

Constraints: $1 \leq N \leq 10^5$; values fit in 32-bit signed integers

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

27. Reverse an Array

Slug: reverse-an-array-027 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Reverse the array in place and print the resulting sequence.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

28. Print Alternate Elements

Slug: print-alternate-elements-028 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print elements at indices 0, 2, 4, and so on.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

29. Find the Second Largest Distinct Value

Slug: find-the-second-largest-distinct-value-029 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print the second largest distinct element, or -1 if it does not exist.

Constraints: $2 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

30. Check if Array Is Sorted

Slug: check-if-array-is-sorted-030 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Determine whether the array is non-decreasing.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

31. Move Zeros to the End

Slug: move-zeros-to-the-end-031 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Move all zero values to the end while preserving the order of non-zero values.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

32. Count Occurrences of a Target

Slug: count-occurrences-of-a-target-032 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many times X occurs in the array.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

33. Find First Occurrence

Slug: find-first-occurrence-033 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print the first index of X, or -1 if absent.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

34. Replace Negative Values

Slug: replace-negative-values-034 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Replace every negative element with zero and print the array.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

35. Difference Between Maximum and Minimum

Slug: difference-between-maximum-and-minimum-035 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print $\max(A) - \min(A)$.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

36. Left Rotate by One

Slug: left-rotate-by-one-036 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate all elements one position to the left.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

37. Right Rotate by One

Slug: right-rotate-by-one-037 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate all elements one position to the right.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

38. Left Rotate by K

Slug: left-rotate-by-k-038 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate the array left by K positions.

Constraints: $1 \leq N \leq 10^5$; $0 \leq K \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

39. Check Majority of Ones

Slug: check-majority-of-ones-039 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Determine whether ones occur more often than zeros in a binary array.

Constraints: $1 \leq N \leq 10^5$; $A[i]$ is 0 or 1

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

40. Find Missing Number

Slug: find-missing-number-040 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: An array contains distinct values from 0 through N with one missing. Find it.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

41. Find the Largest Element

Slug: find-the-largest-element-041 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Given an integer array, print its largest element.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

42. Find the Smallest Element

Slug: find-the-smallest-element-042 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Given an integer array, print its smallest element.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

43. Sum of Array Elements

Slug: sum-of-array-elements-043 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Calculate the sum of all elements in an integer array.

Constraints: $1 \leq N \leq 10^5$; values fit in 32-bit signed integers; sum fits in 64-bit

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

44. Count Positive Values

Slug: count-positive-values-044 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many elements of an array are strictly positive.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

45. Count Negative Values

Slug: count-negative-values-045 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many elements of an array are strictly negative.

Constraints: $1 \leq N \leq 10^5$; $-10^9 \leq A[i] \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

46. Count Even Values

Slug: count-even-values-046 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count the even integers in an array.

Constraints: $1 \leq N \leq 10^5$; values fit in 32-bit signed integers

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

47. Reverse an Array

Slug: reverse-an-array-047 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Reverse the array in place and print the resulting sequence.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

48. Print Alternate Elements

Slug: print-alternate-elements-048 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print elements at indices 0, 2, 4, and so on.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

49. Find the Second Largest Distinct Value

Slug: find-the-second-largest-distinct-value-049 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print the second largest distinct element, or -1 if it does not exist.

Constraints: $2 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

50. Check if Array Is Sorted

Slug: check-if-array-is-sorted-050 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Determine whether the array is non-decreasing.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

51. Move Zeros to the End

Slug: move-zeros-to-the-end-051 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Move all zero values to the end while preserving the order of non-zero values.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

52. Count Occurrences of a Target

Slug: count-occurrences-of-a-target-052 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Count how many times X occurs in the array.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

53. Find First Occurrence

Slug: find-first-occurrence-053 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print the first index of X, or -1 if absent.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

54. Replace Negative Values

Slug: replace-negative-values-054 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Replace every negative element with zero and print the array.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

55. Difference Between Maximum and Minimum

Slug: difference-between-maximum-and-minimum-055 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Print $\max(A) - \min(A)$.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

56. Left Rotate by One

Slug: left-rotate-by-one-056 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate all elements one position to the left.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

57. Right Rotate by One

Slug: right-rotate-by-one-057 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate all elements one position to the right.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

58. Left Rotate by K

Slug: left-rotate-by-k-058 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Rotate the array left by K positions.

Constraints: $1 \leq N \leq 10^5$; $0 \leq K \leq 10^9$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

59. Check Majority of Ones

Slug: check-majority-of-ones-059 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: Determine whether ones occur more often than zeros in a binary array.

Constraints: $1 \leq N \leq 10^5$; $A[i]$ is 0 or 1

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

60. Find Missing Number

Slug: find-missing-number-060 | **Category:** Arrays | **Difficulty:** Easy | **Tags:** arrays, beginner, fundamentals

Problem Statement: An array contains distinct values from 0 through N with one missing. Find it.

Constraints: $1 \leq N \leq 10^5$

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N)

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

61. Reverse a String

Slug: reverse-a-string-061 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse a String' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

62. Count Vowels

Slug: count-vowels-062 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Vowels' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

63. Count Consonants

Slug: count-consonants-063 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Consonants' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

64. Check Palindrome

Slug: check-palindrome-064 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Palindrome' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

65. Convert to Uppercase

Slug: convert-to-uppercase-065 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert to Uppercase' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

66. Convert to Lowercase

Slug: convert-to-lowercase-066 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert to Lowercase' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

67. Remove Spaces

Slug: remove-spaces-067 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Spaces' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

68. Count Words

Slug: count-words-068 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Words' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

69. Count Digits

Slug: count-digits-069 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Digits' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

70. Toggle Letter Case

Slug: toggle-letter-case-070 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Toggle Letter Case' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

71. First Character

Slug: first-character-071 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Character' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

72. Last Character

Slug: last-character-072 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Last Character' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

73. Character Frequency

Slug: character-frequency-073 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Character Frequency' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

74. Remove Vowels

Slug: remove-vowels-074 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Vowels' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

75. Replace Spaces with Hyphens

Slug: replace-spaces-with-hyphens-075 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Replace Spaces with Hyphens' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

76. Compare Two Strings

Slug: compare-two-strings-076 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Compare Two Strings' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

77. Check Anagram

Slug: check-anagram-077 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Anagram' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

78. Count Substring Occurrences

Slug: count-substring-occurrences-078 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Substring Occurrences' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

79. Remove Duplicate Characters

Slug: remove-duplicate-characters-079 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Duplicate Characters' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

80. Find Longest Word

Slug: find-longest-word-080 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Longest Word' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

81. Reverse a String

Slug: reverse-a-string-081 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse a String' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

82. Count Vowels

Slug: count-vowels-082 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Vowels' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

83. Count Consonants

Slug: count-consonants-083 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Consonants' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

84. Check Palindrome

Slug: check-palindrome-084 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Palindrome' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

85. Convert to Uppercase

Slug: convert-to-uppercase-085 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert to Uppercase' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

86. Convert to Lowercase

Slug: convert-to-lowercase-086 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert to Lowercase' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

87. Remove Spaces

Slug: remove-spaces-087 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Spaces' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

88. Count Words

Slug: count-words-088 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Words' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

89. Count Digits

Slug: count-digits-089 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Digits' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

90. Toggle Letter Case

Slug: toggle-letter-case-090 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Toggle Letter Case' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

91. First Character

Slug: first-character-091 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Character' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

92. Last Character

Slug: last-character-092 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Last Character' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

93. Character Frequency

Slug: character-frequency-093 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Character Frequency' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

94. Remove Vowels

Slug: remove-vowels-094 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Vowels' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

95. Replace Spaces with Hyphens

Slug: replace-spaces-with-hyphens-095 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Replace Spaces with Hyphens' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

96. Compare Two Strings

Slug: compare-two-strings-096 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Compare Two Strings' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

97. Check Anagram

Slug: check-anagram-097 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Anagram' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

98. Count Substring Occurrences

Slug: count-substring-occurrences-098 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Substring Occurrences' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

99. Remove Duplicate Characters

Slug: remove-duplicate-characters-099 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Duplicate Characters' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

100. Find Longest Word

Slug: find-longest-word-100 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Longest Word' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

101. Reverse a String

Slug: reverse-a-string-101 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse a String' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

102. Count Vowels

Slug: count-vowels-102 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Vowels' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

103. Count Consonants

Slug: count-consonants-103 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Consonants' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

104. Check Palindrome

Slug: check-palindrome-104 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Palindrome' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

105. Convert to Uppercase

Slug: convert-to-uppercase-105 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert to Uppercase' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

106. Convert to Lowercase

Slug: convert-to-lowercase-106 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert to Lowercase' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

107. Remove Spaces

Slug: remove-spaces-107 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Spaces' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

108. Count Words

Slug: count-words-108 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Words' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

109. Count Digits

Slug: count-digits-109 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Digits' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

110. Toggle Letter Case

Slug: toggle-letter-case-110 | **Category:** Strings | **Difficulty:** Easy | **Tags:** strings, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Toggle Letter Case' using the strings topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

111. Linear Search

Slug: linear-search-111 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Linear Search' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

112. First Occurrence in Sorted Array

Slug: first-occurrence-in-sorted-array-112 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Occurrence in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

113. Last Occurrence in Sorted Array

Slug: last-occurrence-in-sorted-array-113 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Last Occurrence in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

114. Binary Search

Slug: binary-search-114 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Binary Search' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

115. Count Target in Sorted Array

Slug: count-target-in-sorted-array-115 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Target in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

116. Search Insert Position

Slug: search-insert-position-116 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Insert Position' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

117. Find Floor

Slug: find-floor-117 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Floor' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

118. Find Ceiling

Slug: find-ceiling-118 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Ceiling' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

119. Find Square Root

Slug: find-square-root-119 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Square Root' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

120. Find Peak in Mountain Array

Slug: find-peak-in-mountain-array-120 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Peak in Mountain Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

121. Linear Search

Slug: linear-search-121 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Linear Search' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

122. First Occurrence in Sorted Array

Slug: first-occurrence-in-sorted-array-122 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Occurrence in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

123. Last Occurrence in Sorted Array

Slug: last-occurrence-in-sorted-array-123 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Last Occurrence in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

124. Binary Search

Slug: binary-search-124 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Binary Search' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

125. Count Target in Sorted Array

Slug: count-target-in-sorted-array-125 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Target in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

126. Search Insert Position

Slug: search-insert-position-126 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Insert Position' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

127. Find Floor

Slug: find-floor-127 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Floor' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

128. Find Ceiling

Slug: find-ceiling-128 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Ceiling' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

129. Find Square Root

Slug: find-square-root-129 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Square Root' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

130. Find Peak in Mountain Array

Slug: find-peak-in-mountain-array-130 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Peak in Mountain Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

131. Linear Search

Slug: linear-search-131 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Linear Search' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

132. First Occurrence in Sorted Array

Slug: first-occurrence-in-sorted-array-132 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Occurrence in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

133. Last Occurrence in Sorted Array

Slug: last-occurrence-in-sorted-array-133 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Last Occurrence in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

134. Binary Search

Slug: binary-search-134 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Binary Search' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

135. Count Target in Sorted Array

Slug: count-target-in-sorted-array-135 | **Category:** Searching | **Difficulty:** Easy | **Tags:** searching, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Target in Sorted Array' using the searching topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

136. Bubble Sort

Slug: bubble-sort-136 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Bubble Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

137. Selection Sort

Slug: selection-sort-137 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Selection Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

138. Insertion Sort

Slug: insertion-sort-138 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insertion Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

139. Sort 0s and 1s

Slug: sort-0s-and-1s-139 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort 0s and 1s' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

140. Merge Two Sorted Arrays

Slug: merge-two-sorted-arrays-140 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Merge Two Sorted Arrays' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

141. Sort by Absolute Value

Slug: sort-by-absolute-value-141 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort by Absolute Value' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

142. Sort Characters

Slug: sort-characters-142 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort Characters' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

143. Sort Array Ascending

Slug: sort-array-ascending-143 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort Array Ascending' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

144. Sort Array Descending

Slug: sort-array-descending-144 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort Array Descending' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

145. Check Nearly Sorted Array

Slug: check-nearly-sorted-array-145 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Nearly Sorted Array' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

146. Bubble Sort

Slug: bubble-sort-146 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Bubble Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

147. Selection Sort

Slug: selection-sort-147 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Selection Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

148. Insertion Sort

Slug: insertion-sort-148 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insertion Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

149. Sort 0s and 1s

Slug: sort-0s-and-1s-149 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort 0s and 1s' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

150. Merge Two Sorted Arrays

Slug: merge-two-sorted-arrays-150 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Merge Two Sorted Arrays' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

151. Sort by Absolute Value

Slug: sort-by-absolute-value-151 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort by Absolute Value' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

152. Sort Characters

Slug: sort-characters-152 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort Characters' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

153. Sort Array Ascending

Slug: sort-array-ascending-153 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort Array Ascending' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

154. Sort Array Descending

Slug: sort-array-descending-154 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort Array Descending' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

155. Check Nearly Sorted Array

Slug: check-nearly-sorted-array-155 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Nearly Sorted Array' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

156. Bubble Sort

Slug: bubble-sort-156 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Bubble Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

157. Selection Sort

Slug: selection-sort-157 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Selection Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

158. Insertion Sort

Slug: insertion-sort-158 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insertion Sort' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

159. Sort 0s and 1s

Slug: sort-0s-and-1s-159 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sort 0s and 1s' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

160. Merge Two Sorted Arrays

Slug: merge-two-sorted-arrays-160 | **Category:** Sorting | **Difficulty:** Easy | **Tags:** sorting, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Merge Two Sorted Arrays' using the sorting topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

161. Frequency Map

Slug: frequency-map-161 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Frequency Map' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

162. First Repeating Element

Slug: first-repeating-element-162 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Repeating Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

163. First Non-Repeating Element

Slug: first-non-repeating-element-163 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Non-Repeating Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

164. Contains Duplicate

Slug: contains-duplicate-164 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Contains Duplicate' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

165. Unique Element

Slug: unique-element-165 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Unique Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

166. Common Elements

Slug: common-elements-166 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Common Elements' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

167. Distinct Count

Slug: distinct-count-167 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Distinct Count' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

168. Intersection Size

Slug: intersection-size-168 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Intersection Size' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

169. Most Frequent Value

Slug: most-frequent-value-169 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Most Frequent Value' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

170. Pair Sum Exists

Slug: pair-sum-exists-170 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Pair Sum Exists' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

171. Frequency Map

Slug: frequency-map-171 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Frequency Map' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

172. First Repeating Element

Slug: first-repeating-element-172 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Repeating Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

173. First Non-Repeating Element

Slug: first-non-repeating-element-173 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Non-Repeating Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

174. Contains Duplicate

Slug: contains-duplicate-174 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Contains Duplicate' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

175. Unique Element

Slug: unique-element-175 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Unique Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

176. Common Elements

Slug: common-elements-176 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Common Elements' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

177. Distinct Count

Slug: distinct-count-177 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Distinct Count' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

178. Intersection Size

Slug: intersection-size-178 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Intersection Size' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

179. Most Frequent Value

Slug: most-frequent-value-179 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Most Frequent Value' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

180. Pair Sum Exists

Slug: pair-sum-exists-180 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Pair Sum Exists' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

181. Frequency Map

Slug: frequency-map-181 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Frequency Map' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

182. First Repeating Element

Slug: first-repeating-element-182 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Repeating Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

183. First Non-Repeating Element

Slug: first-non-repeating-element-183 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Non-Repeating Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

184. Contains Duplicate

Slug: contains-duplicate-184 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Contains Duplicate' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

185. Unique Element

Slug: unique-element-185 | **Category:** Hashing | **Difficulty:** Easy | **Tags:** hashing, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Unique Element' using the hashing topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

186. Pair With Target Sum

Slug: pair-with-target-sum-186 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Pair With Target Sum' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

187. Reverse Vowels

Slug: reverse-vowels-187 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Vowels' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

188. Remove Duplicates from Sorted Array

Slug: remove-duplicates-from-sorted-array-188 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Duplicates from Sorted Array' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

189. Move Negatives to One Side

Slug: move-negatives-to-one-side-189 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Move Negatives to One Side' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

190. Maximum Sum Window of Size K

Slug: maximum-sum-window-of-size-k-190 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Sum Window of Size K' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

191. Minimum Sum Window of Size K

Slug: minimum-sum-window-of-size-k-191 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Sum Window of Size K' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

192. Longest Run of Ones

Slug: longest-run-of-ones-192 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Longest Run of Ones' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

193. Check Palindrome with Two Pointers

Slug: check-palindrome-with-two-pointers-193 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Palindrome with Two Pointers' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

194. Pair With Target Sum

Slug: pair-with-target-sum-194 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Pair With Target Sum' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

195. Reverse Vowels

Slug: reverse-vowels-195 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Vowels' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

196. Remove Duplicates from Sorted Array

Slug: remove-duplicates-from-sorted-array-196 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Duplicates from Sorted Array' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

197. Move Negatives to One Side

Slug: move-negatives-to-one-side-197 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Move Negatives to One Side' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

198. Maximum Sum Window of Size K

Slug: maximum-sum-window-of-size-k-198 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Sum Window of Size K' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

199. Minimum Sum Window of Size K

Slug: minimum-sum-window-of-size-k-199 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Sum Window of Size K' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

200. Longest Run of Ones

Slug: longest-run-of-ones-200 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Longest Run of Ones' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

201. Check Palindrome with Two Pointers

Slug: check-palindrome-with-two-pointers-201 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Palindrome with Two Pointers' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

202. Pair With Target Sum

Slug: pair-with-target-sum-202 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Pair With Target Sum' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

203. Reverse Vowels

Slug: reverse-vowels-203 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Vowels' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

204. Remove Duplicates from Sorted Array

Slug: remove-duplicates-from-sorted-array-204 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Duplicates from Sorted Array' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

205. Move Negatives to One Side

Slug: move-negatives-to-one-side-205 | **Category:** Two Pointers & Sliding Window | **Difficulty:** Easy | **Tags:** two pointers & sliding window, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Move Negatives to One Side' using the two pointers & sliding window topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

206. Prime Check

Slug: prime-check-206 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Prime Check' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

207. Print Primes up to N

Slug: print-primes-up-to-n-207 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Print Primes up to N' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

208. Factorial

Slug: factorial-208 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Factorial' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

209. Fibonacci Number

Slug: fibonacci-number-209 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

210. GCD

Slug: gcd-210 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'GCD' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

211. LCM

Slug: lcm-211 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'LCM' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

212. Reverse Number

Slug: reverse-number-212 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

213. Palindrome Number

Slug: palindrome-number-213 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Palindrome Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

214. Armstrong Number

Slug: armstrong-number-214 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Armstrong Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

215. Perfect Number

Slug: perfect-number-215 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Perfect Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

216. Digit Sum

Slug: digit-sum-216 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Digit Sum' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

217. Digit Product

Slug: digit-product-217 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Digit Product' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

218. Count Digits

Slug: count-digits-218 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Digits' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

219. Power of Two

Slug: power-of-two-219 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Power of Two' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

220. Leap Year

Slug: leap-year-220 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Leap Year' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

221. Prime Check

Slug: prime-check-221 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Prime Check' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

222. Print Primes up to N

Slug: print-primes-up-to-n-222 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Print Primes up to N' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

223. Factorial

Slug: factorial-223 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Factorial' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

224. Fibonacci Number

Slug: fibonacci-number-224 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

225. GCD

Slug: gcd-225 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'GCD' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

226. LCM

Slug: lcm-226 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'LCM' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

227. Reverse Number

Slug: reverse-number-227 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

228. Palindrome Number

Slug: palindrome-number-228 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Palindrome Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

229. Armstrong Number

Slug: armstrong-number-229 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Armstrong Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

230. Perfect Number

Slug: perfect-number-230 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Perfect Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

231. Digit Sum

Slug: digit-sum-231 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Digit Sum' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

232. Digit Product

Slug: digit-product-232 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Digit Product' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

233. Count Digits

Slug: count-digits-233 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Digits' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

234. Power of Two

Slug: power-of-two-234 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Power of Two' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

235. Leap Year

Slug: leap-year-235 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Leap Year' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

236. Prime Check

Slug: prime-check-236 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Prime Check' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

237. Print Primes up to N

Slug: print-primes-up-to-n-237 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Print Primes up to N' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

238. Factorial

Slug: factorial-238 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Factorial' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

239. Fibonacci Number

Slug: fibonacci-number-239 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci Number' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

240. GCD

Slug: gcd-240 | **Category:** Math & Number Theory | **Difficulty:** Easy | **Tags:** math & number theory, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'GCD' using the math & number theory topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

241. Print 1 to N

Slug: print-1-to-n-241 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Print 1 to N' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

242. Print N to 1

Slug: print-n-to-1-242 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Print N to 1' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

243. Recursive Sum

Slug: recursive-sum-243 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Sum' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

244. Recursive Factorial

Slug: recursive-factorial-244 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Factorial' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

245. Recursive Power

Slug: recursive-power-245 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Power' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

246. Recursive Digit Sum

Slug: recursive-digit-sum-246 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Digit Sum' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

247. Recursive String Reverse

Slug: recursive-string-reverse-247 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive String Reverse' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

248. Recursive Palindrome

Slug: recursive-palindrome-248 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Palindrome' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

249. Count Zeros

Slug: count-zeros-249 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Zeros' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

250. Fibonacci Recursion

Slug: fibonacci-recursion-250 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci Recursion' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

251. Print 1 to N

Slug: print-1-to-n-251 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Print 1 to N' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

252. Print N to 1

Slug: print-n-to-1-252 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Print N to 1' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

253. Recursive Sum

Slug: recursive-sum-253 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Sum' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

254. Recursive Factorial

Slug: recursive-factorial-254 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Factorial' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

255. Recursive Power

Slug: recursive-power-255 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Power' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

256. Recursive Digit Sum

Slug: recursive-digit-sum-256 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Digit Sum' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

257. Recursive String Reverse

Slug: recursive-string-reverse-257 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive String Reverse' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

258. Recursive Palindrome

Slug: recursive-palindrome-258 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Recursive Palindrome' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

259. Count Zeros

Slug: count-zeros-259 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Zeros' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

260. Fibonacci Recursion

Slug: fibonacci-recursion-260 | **Category:** Recursion | **Difficulty:** Easy | **Tags:** recursion, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci Recursion' using the recursion topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

261. Count Nodes

Slug: count-nodes-261 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Nodes' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

262. Search Node

Slug: search-node-262 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

263. Insert at Head

Slug: insert-at-head-263 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert at Head' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

264. Insert at Tail

Slug: insert-at-tail-264 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert at Tail' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

265. Delete Head

Slug: delete-head-265 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete Head' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

266. Delete Tail

Slug: delete-tail-266 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete Tail' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

267. Delete First Matching Node

Slug: delete-first-matching-node-267 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete First Matching Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

268. Reverse List

Slug: reverse-list-268 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse List' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

269. Find Middle Node

Slug: find-middle-node-269 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Middle Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

270. Find Length

Slug: find-length-270 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Length' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

271. Compare Two Lists

Slug: compare-two-lists-271 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Compare Two Lists' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

272. Merge Two Sorted Lists

Slug: merge-two-sorted-lists-272 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Merge Two Sorted Lists' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

273. Count Nodes

Slug: count-nodes-273 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Nodes' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

274. Search Node

Slug: search-node-274 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

275. Insert at Head

Slug: insert-at-head-275 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert at Head' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

276. Insert at Tail

Slug: insert-at-tail-276 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert at Tail' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

277. Delete Head

Slug: delete-head-277 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete Head' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

278. Delete Tail

Slug: delete-tail-278 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete Tail' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

279. Delete First Matching Node

Slug: delete-first-matching-node-279 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete First Matching Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

280. Reverse List

Slug: reverse-list-280 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse List' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

281. Find Middle Node

Slug: find-middle-node-281 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Middle Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

282. Find Length

Slug: find-length-282 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Length' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

283. Compare Two Lists

Slug: compare-two-lists-283 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Compare Two Lists' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

284. Merge Two Sorted Lists

Slug: merge-two-sorted-lists-284 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Merge Two Sorted Lists' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

285. Count Nodes

Slug: count-nodes-285 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Nodes' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

286. Search Node

Slug: search-node-286 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

287. Insert at Head

Slug: insert-at-head-287 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert at Head' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

288. Insert at Tail

Slug: insert-at-tail-288 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert at Tail' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

289. Delete Head

Slug: delete-head-289 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete Head' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

290. Delete Tail

Slug: delete-tail-290 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete Tail' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

291. Delete First Matching Node

Slug: delete-first-matching-node-291 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Delete First Matching Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

292. Reverse List

Slug: reverse-list-292 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse List' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

293. Find Middle Node

Slug: find-middle-node-293 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Middle Node' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

294. Find Length

Slug: find-length-294 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Length' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

295. Compare Two Lists

Slug: compare-two-lists-295 | **Category:** Linked Lists | **Difficulty:** Easy | **Tags:** linked lists, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Compare Two Lists' using the linked lists topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

296. Stack Operations

Slug: stack-operations-296 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Stack Operations' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

297. Balanced Parentheses

Slug: balanced-parentheses-297 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Balanced Parentheses' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

298. Reverse String with Stack

Slug: reverse-string-with-stack-298 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse String with Stack' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

299. Remove Adjacent Duplicates

Slug: remove-adjacent-duplicates-299 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Adjacent Duplicates' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

300. Next Greater Element

Slug: next-greater-element-300 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Next Greater Element' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

301. Min Stack

Slug: min-stack-301 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Stack' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

302. Evaluate Simple Postfix

Slug: evaluate-simple-postfix-302 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Evaluate Simple Postfix' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

303. Convert Decimal to Binary

Slug: convert-decimal-to-binary-303 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert Decimal to Binary' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

304. Stack Operations

Slug: stack-operations-304 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Stack Operations' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

305. Balanced Parentheses

Slug: balanced-parentheses-305 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Balanced Parentheses' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

306. Reverse String with Stack

Slug: reverse-string-with-stack-306 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse String with Stack' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

307. Remove Adjacent Duplicates

Slug: remove-adjacent-duplicates-307 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Adjacent Duplicates' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

308. Next Greater Element

Slug: next-greater-element-308 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Next Greater Element' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

309. Min Stack

Slug: min-stack-309 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Stack' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

310. Evaluate Simple Postfix

Slug: evaluate-simple-postfix-310 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Evaluate Simple Postfix' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

311. Convert Decimal to Binary

Slug: convert-decimal-to-binary-311 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert Decimal to Binary' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

312. Stack Operations

Slug: stack-operations-312 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Stack Operations' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

313. Balanced Parentheses

Slug: balanced-parentheses-313 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Balanced Parentheses' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

314. Reverse String with Stack

Slug: reverse-string-with-stack-314 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse String with Stack' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

315. Remove Adjacent Duplicates

Slug: remove-adjacent-duplicates-315 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Adjacent Duplicates' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

316. Next Greater Element

Slug: next-greater-element-316 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Next Greater Element' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

317. Min Stack

Slug: min-stack-317 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Stack' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

318. Evaluate Simple Postfix

Slug: evaluate-simple-postfix-318 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Evaluate Simple Postfix' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

319. Convert Decimal to Binary

Slug: convert-decimal-to-binary-319 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Convert Decimal to Binary' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

320. Stack Operations

Slug: stack-operations-320 | **Category:** Stacks | **Difficulty:** Easy | **Tags:** stacks, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Stack Operations' using the stacks topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

321. Queue Operations

Slug: queue-operations-321 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Queue Operations' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

322. Circular Queue

Slug: circular-queue-322 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Circular Queue' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

323. Reverse Queue

Slug: reverse-queue-323 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Queue' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

324. Generate Binary Numbers

Slug: generate-binary-numbers-324 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Generate Binary Numbers' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

325. First Negative in Window

Slug: first-negative-in-window-325 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Negative in Window' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

326. Queue Using Two Stacks

Slug: queue-using-two-stacks-326 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Queue Using Two Stacks' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

327. Stack Using Two Queues

Slug: stack-using-two-queues-327 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Stack Using Two Queues' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

328. Queue Operations

Slug: queue-operations-328 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Queue Operations' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

329. Circular Queue

Slug: circular-queue-329 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Circular Queue' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

330. Reverse Queue

Slug: reverse-queue-330 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Queue' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

331. Generate Binary Numbers

Slug: generate-binary-numbers-331 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Generate Binary Numbers' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

332. First Negative in Window

Slug: first-negative-in-window-332 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Negative in Window' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

333. Queue Using Two Stacks

Slug: queue-using-two-stacks-333 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Queue Using Two Stacks' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

334. Stack Using Two Queues

Slug: stack-using-two-queues-334 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Stack Using Two Queues' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

335. Queue Operations

Slug: queue-operations-335 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Queue Operations' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

336. Circular Queue

Slug: circular-queue-336 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Circular Queue' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

337. Reverse Queue

Slug: reverse-queue-337 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Reverse Queue' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

338. Generate Binary Numbers

Slug: generate-binary-numbers-338 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Generate Binary Numbers' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

339. First Negative in Window

Slug: first-negative-in-window-339 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'First Negative in Window' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

340. Queue Using Two Stacks

Slug: queue-using-two-stacks-340 | **Category:** Queues | **Difficulty:** Easy | **Tags:** queues, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Queue Using Two Stacks' using the queues topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

341. Preorder Traversal

Slug: preorder-traversal-341 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Preorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

342. Inorder Traversal

Slug: inorder-traversal-342 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Inorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

343. Postorder Traversal

Slug: postorder-traversal-343 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Postorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

344. Level Order Traversal

Slug: level-order-traversal-344 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Level Order Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

345. Count Nodes

Slug: count-nodes-345 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

346. Count Leaves

Slug: count-leaves-346 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Leaves' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

347. Tree Height

Slug: tree-height-347 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Tree Height' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

348. Sum of Nodes

Slug: sum-of-nodes-348 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sum of Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

349. Maximum Node

Slug: maximum-node-349 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Node' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

350. Search Value

Slug: search-value-350 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Value' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

351. Mirror Tree

Slug: mirror-tree-351 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Mirror Tree' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

352. Count Full Nodes

Slug: count-full-nodes-352 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Full Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

353. Preorder Traversal

Slug: preorder-traversal-353 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Preorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

354. Inorder Traversal

Slug: inorder-traversal-354 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Inorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

355. Postorder Traversal

Slug: postorder-traversal-355 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Postorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

356. Level Order Traversal

Slug: level-order-traversal-356 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Level Order Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

357. Count Nodes

Slug: count-nodes-357 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

358. Count Leaves

Slug: count-leaves-358 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Leaves' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

359. Tree Height

Slug: tree-height-359 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Tree Height' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

360. Sum of Nodes

Slug: sum-of-nodes-360 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sum of Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

361. Maximum Node

Slug: maximum-node-361 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Node' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

362. Search Value

Slug: search-value-362 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Value' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

363. Mirror Tree

Slug: mirror-tree-363 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Mirror Tree' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

364. Count Full Nodes

Slug: count-full-nodes-364 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Full Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

365. Preorder Traversal

Slug: preorder-traversal-365 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Preorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

366. Inorder Traversal

Slug: inorder-traversal-366 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Inorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

367. Postorder Traversal

Slug: postorder-traversal-367 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Postorder Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

368. Level Order Traversal

Slug: level-order-traversal-368 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Level Order Traversal' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

369. Count Nodes

Slug: count-nodes-369 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

370. Count Leaves

Slug: count-leaves-370 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Leaves' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

371. Tree Height

Slug: tree-height-371 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Tree Height' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

372. Sum of Nodes

Slug: sum-of-nodes-372 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Sum of Nodes' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

373. Maximum Node

Slug: maximum-node-373 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Node' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

374. Search Value

Slug: search-value-374 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search Value' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

375. Mirror Tree

Slug: mirror-tree-375 | **Category:** Binary Trees | **Difficulty:** Easy | **Tags:** binary trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Mirror Tree' using the binary trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

376. Search BST

Slug: search-bst-376 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

377. Insert BST

Slug: insert-bst-377 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

378. Minimum BST Value

Slug: minimum-bst-value-378 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum BST Value' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

379. Maximum BST Value

Slug: maximum-bst-value-379 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum BST Value' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

380. Validate BST

Slug: validate-bst-380 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Validate BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

381. Find Inorder Successor

Slug: find-inorder-successor-381 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Inorder Successor' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

382. Search BST

Slug: search-bst-382 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

383. Insert BST

Slug: insert-bst-383 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

384. Minimum BST Value

Slug: minimum-bst-value-384 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum BST Value' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

385. Maximum BST Value

Slug: maximum-bst-value-385 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum BST Value' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

386. Validate BST

Slug: validate-bst-386 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Validate BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

387. Find Inorder Successor

Slug: find-inorder-successor-387 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Find Inorder Successor' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

388. Search BST

Slug: search-bst-388 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

389. Insert BST

Slug: insert-bst-389 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Insert BST' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

390. Minimum BST Value

Slug: minimum-bst-value-390 | **Category:** Binary Search Trees | **Difficulty:** Easy | **Tags:** binary search trees, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum BST Value' using the binary search trees topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

391. Kth Largest in Small Array

Slug: kth-largest-in-small-array-391 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Kth Largest in Small Array' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

392. Kth Smallest in Small Array

Slug: kth-smallest-in-small-array-392 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Kth Smallest in Small Array' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

393. Min Heap Insert

Slug: min-heap-insert-393 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Heap Insert' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

394. Remove Minimum

Slug: remove-minimum-394 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Minimum' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

395. Heap Sort

Slug: heap-sort-395 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Heap Sort' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

396. Top K Smallest Values

Slug: top-k-smallest-values-396 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Top K Smallest Values' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

397. Kth Largest in Small Array

Slug: kth-largest-in-small-array-397 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Kth Largest in Small Array' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

398. Kth Smallest in Small Array

Slug: kth-smallest-in-small-array-398 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Kth Smallest in Small Array' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

399. Min Heap Insert

Slug: min-heap-insert-399 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Heap Insert' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

400. Remove Minimum

Slug: remove-minimum-400 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Remove Minimum' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

401. Heap Sort

Slug: heap-sort-401 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Heap Sort' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

402. Top K Smallest Values

Slug: top-k-smallest-values-402 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Top K Smallest Values' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

403. Kth Largest in Small Array

Slug: kth-largest-in-small-array-403 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Kth Largest in Small Array' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

404. Kth Smallest in Small Array

Slug: kth-smallest-in-small-array-404 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Kth Smallest in Small Array' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

405. Min Heap Insert

Slug: min-heap-insert-405 | **Category:** Heaps | **Difficulty:** Easy | **Tags:** heaps, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Heap Insert' using the heaps topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

406. BFS Traversal

Slug: bfs-traversal-406 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'BFS Traversal' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

407. DFS Traversal

Slug: dfs-traversal-407 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'DFS Traversal' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

408. Count Connected Components

Slug: count-connected-components-408 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Connected Components' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

409. Check Path Exists

Slug: check-path-exists-409 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Path Exists' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

410. Count Degree

Slug: count-degree-410 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Degree' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

411. Undirected Cycle Check

Slug: undirected-cycle-check-411 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Undirected Cycle Check' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

412. Grid Flood Fill

Slug: grid-flood-fill-412 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Grid Flood Fill' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

413. Number of Islands

Slug: number-of-islands-413 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Number of Islands' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

414. BFS Traversal

Slug: bfs-traversal-414 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'BFS Traversal' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

415. DFS Traversal

Slug: dfs-traversal-415 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'DFS Traversal' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

416. Count Connected Components

Slug: count-connected-components-416 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Connected Components' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

417. Check Path Exists

Slug: check-path-exists-417 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Path Exists' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

418. Count Degree

Slug: count-degree-418 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Degree' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

419. Undirected Cycle Check

Slug: undirected-cycle-check-419 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Undirected Cycle Check' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

420. Grid Flood Fill

Slug: grid-flood-fill-420 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Grid Flood Fill' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

421. Number of Islands

Slug: number-of-islands-421 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Number of Islands' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

422. BFS Traversal

Slug: bfs-traversal-422 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'BFS Traversal' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

423. DFS Traversal

Slug: dfs-traversal-423 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'DFS Traversal' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

424. Count Connected Components

Slug: count-connected-components-424 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Connected Components' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

425. Check Path Exists

Slug: check-path-exists-425 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Path Exists' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

426. Count Degree

Slug: count-degree-426 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Degree' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

427. Undirected Cycle Check

Slug: undirected-cycle-check-427 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Undirected Cycle Check' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

428. Grid Flood Fill

Slug: grid-flood-fill-428 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Grid Flood Fill' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

429. Number of Islands

Slug: number-of-islands-429 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Number of Islands' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

430. BFS Traversal

Slug: bfs-traversal-430 | **Category:** Graphs | **Difficulty:** Easy | **Tags:** graphs, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'BFS Traversal' using the graphs topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

431. Assign Cookies

Slug: assign-cookies-431 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Assign Cookies' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

432. Minimum Coins for Standard Denominations

Slug: minimum-coins-for-standard-denominations-432 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Coins for Standard Denominations' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

433. Activity Selection

Slug: activity-selection-433 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Activity Selection' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

434. Minimum Platforms

Slug: minimum-platforms-434 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Platforms' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

435. Fractional Knapsack Small Case

Slug: fractional-knapsack-small-case-435 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fractional Knapsack Small Case' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

436. Assign Cookies

Slug: assign-cookies-436 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Assign Cookies' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

437. Minimum Coins for Standard Denominations

Slug: minimum-coins-for-standard-denominations-437 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Coins for Standard Denominations' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

438. Activity Selection

Slug: activity-selection-438 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Activity Selection' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

439. Minimum Platforms

Slug: minimum-platforms-439 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Platforms' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

440. Fractional Knapsack Small Case

Slug: fractional-knapsack-small-case-440 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fractional Knapsack Small Case' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

441. Assign Cookies

Slug: assign-cookies-441 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Assign Cookies' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

442. Minimum Coins for Standard Denominations

Slug: minimum-coins-for-standard-denominations-442 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Coins for Standard Denominations' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

443. Activity Selection

Slug: activity-selection-443 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Activity Selection' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

444. Minimum Platforms

Slug: minimum-platforms-444 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Platforms' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

445. Fractional Knapsack Small Case

Slug: fractional-knapsack-small-case-445 | **Category:** Greedy | **Difficulty:** Easy | **Tags:** greedy, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fractional Knapsack Small Case' using the greedy topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

446. Fibonacci DP

Slug: fibonacci-dp-446 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci DP' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

447. Climbing Stairs

Slug: climbing-stairs-447 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Climbing Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

448. Minimum Cost Stairs

Slug: minimum-cost-stairs-448 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Cost Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

449. House Robber

Slug: house-robber-449 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'House Robber' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

450. Coin Change Ways

Slug: coin-change-ways-450 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Coin Change Ways' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

451. Unique Grid Paths

Slug: unique-grid-paths-451 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Unique Grid Paths' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

452. Grid Paths with Obstacles

Slug: grid-paths-with-obstacles-452 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Grid Paths with Obstacles' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

453. Maximum Sum Non-Adjacent

Slug: maximum-sum-non-adjacent-453 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Sum Non-Adjacent' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

454. Min Steps to One

Slug: min-steps-to-one-454 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Steps to One' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

455. Fibonacci DP

Slug: fibonacci-dp-455 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci DP' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

456. Climbing Stairs

Slug: climbing-stairs-456 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Climbing Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

457. Minimum Cost Stairs

Slug: minimum-cost-stairs-457 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Cost Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

458. House Robber

Slug: house-robber-458 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'House Robber' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

459. Coin Change Ways

Slug: coin-change-ways-459 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Coin Change Ways' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

460. Unique Grid Paths

Slug: unique-grid-paths-460 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Unique Grid Paths' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

461. Grid Paths with Obstacles

Slug: grid-paths-with-obstacles-461 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Grid Paths with Obstacles' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

462. Maximum Sum Non-Adjacent

Slug: maximum-sum-non-adjacent-462 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Sum Non-Adjacent' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

463. Min Steps to One

Slug: min-steps-to-one-463 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Steps to One' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

464. Fibonacci DP

Slug: fibonacci-dp-464 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci DP' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

465. Climbing Stairs

Slug: climbing-stairs-465 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Climbing Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

466. Minimum Cost Stairs

Slug: minimum-cost-stairs-466 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Cost Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

467. House Robber

Slug: house-robber-467 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'House Robber' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

468. Coin Change Ways

Slug: coin-change-ways-468 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Coin Change Ways' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

469. Unique Grid Paths

Slug: unique-grid-paths-469 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Unique Grid Paths' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

470. Grid Paths with Obstacles

Slug: grid-paths-with-obstacles-470 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Grid Paths with Obstacles' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

471. Maximum Sum Non-Adjacent

Slug: maximum-sum-non-adjacent-471 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Maximum Sum Non-Adjacent' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

472. Min Steps to One

Slug: min-steps-to-one-472 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Min Steps to One' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

473. Fibonacci DP

Slug: fibonacci-dp-473 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Fibonacci DP' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

474. Climbing Stairs

Slug: climbing-stairs-474 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Climbing Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

475. Minimum Cost Stairs

Slug: minimum-cost-stairs-475 | **Category:** Dynamic Programming | **Difficulty:** Easy | **Tags:** dynamic programming, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Minimum Cost Stairs' using the dynamic programming topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

476. Check Odd or Even with Bits

Slug: check-odd-or-even-with-bits-476 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Odd or Even with Bits' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

477. Count Set Bits

Slug: count-set-bits-477 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Set Bits' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

478. Power of Two with Bits

Slug: power-of-two-with-bits-478 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Power of Two with Bits' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

479. Single Number

Slug: single-number-479 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Single Number' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

480. Toggle Bit

Slug: toggle-bit-480 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Toggle Bit' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

481. Set Bit

Slug: set-bit-481 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Set Bit' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

482. Clear Bit

Slug: clear-bit-482 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Clear Bit' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

483. XOR from 1 to N

Slug: xor-from-1-to-n-483 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'XOR from 1 to N' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

484. Check Odd or Even with Bits

Slug: check-odd-or-even-with-bits-484 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Check Odd or Even with Bits' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

485. Count Set Bits

Slug: count-set-bits-485 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Count Set Bits' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

486. Power of Two with Bits

Slug: power-of-two-with-bits-486 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Power of Two with Bits' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

487. Single Number

Slug: single-number-487 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Single Number' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

488. Toggle Bit

Slug: toggle-bit-488 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Toggle Bit' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

489. Set Bit

Slug: set-bit-489 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Set Bit' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

490. Clear Bit

Slug: clear-bit-490 | **Category:** Bit Manipulation | **Difficulty:** Easy | **Tags:** bit manipulation, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Clear Bit' using the bit manipulation topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

491. Matrix Transpose

Slug: matrix-transpose-491 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Matrix Transpose' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

492. Matrix Diagonal Sum

Slug: matrix-diagonal-sum-492 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Matrix Diagonal Sum' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

493. Row with Maximum Sum

Slug: row-with-maximum-sum-493 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Row with Maximum Sum' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N . The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

494. Column with Maximum Sum

Slug: column-with-maximum-sum-494 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Column with Maximum Sum' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

495. Spiral Print

Slug: spiral-print-495 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Spiral Print' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

496. Search in Matrix

Slug: search-in-matrix-496 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Search in Matrix' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

497. Rotate Matrix 90 Degrees

Slug: rotate-matrix-90-degrees-497 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Rotate Matrix 90 Degrees' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0
Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9
Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7
Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

498. Matrix Transpose

Slug: matrix-transpose-498 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Matrix Transpose' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5
Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3
Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

499. Matrix Diagonal Sum

Slug: matrix-diagonal-sum-499 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Matrix Diagonal Sum' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: O(N) target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.

500. Row with Maximum Sum

Slug: row-with-maximum-sum-500 | **Category:** Matrices | **Difficulty:** Easy | **Tags:** matrices, beginner, fundamentals

Problem Statement: Solve the beginner-level problem 'Row with Maximum Sum' using the matrices topic. Read the provided input, perform the required operation, and print the exact result.

Constraints: $1 \leq N \leq 10^5$ unless the statement specifies a smaller structure.

Input Format: The first line contains N. The second line contains N space-separated integers, unless the problem requires a string, matrix, linked list, tree, or graph representation as described in the statement.

Output Format: Print the required answer in the specified format. Do not print labels or extra text.

Visible Test 1: Input: 5 | 1 2 3 4 5

Expected Output: 5

Visible Test 2: Input: 6 | -4 0 7 7 -2 3

Expected Output: 7

Visible Test 3: Input: 1 | 0

Expected Output: 0

Hidden Test 1: Input: 8 | 9 9 9 9 9 9 9 9

Expected Output: 9

Hidden Test 2: Input: 7 | -1000000000 -5 -9 -2 -8 -3 -7

Expected Output: -2

Hidden Test 3: Maximum-size randomized and boundary-value case; expected output must be generated by the reference solution and stored server-side.

Complexity Target: $O(N)$ target unless the task naturally requires sorting or traversal.

Metadata: unique_id, slug, title, category, difficulty, statement, constraints, input_format, output_format, visible_tests, hidden_tests, tags, complexity_target.